

OH-XTREME®

ENSURES A LONG USEFUL LIFE.

MECHANICAL COMPONENTS OF INJECTION MOULDS ARE EXPOSED TO VERY HIGH STRAINS. OTTO HOFSTETTER AG HAS DEVELOPED OH-XTREME® TO REDUCE MOULD WEAR SUBSTANTIALLY. OH-XTREME® HAS BEEN ENSURING TROUBLE-FREE OPERATION AND A LONG USEFUL LIFE OF THE INJECTION MOULDS FROM UZNACH FOR MORE THAN 10 YEARS.

Thread slides coated
with OH-Xtreme®.



The first generation of moulds from Otto Hofstetter AG was manufactured in the early 1960s. It was standard practice in those days to rely on lubricants such as oil or grease to minimise wear of excessively strained, hardened surfaces. In spite of this protective measure, the surfaces in question were damaged very quickly, which is why they were provided with a chromium layer. While this reduced wear markedly, it had no effect at all on the level of maintenance required.

Golden times.

New coatings became available towards the end of the 1990s. They not only promised better properties in terms of abrasive wear, but also had a positive influence on the ejection of products from the injection moulds. The TiN (titanium nitride) coating, for example, which was characterised above all by its attractive golden shine. This coating was also applied in the moulds from Otto Hofstetter AG for a long time, but a relatively high maintenance was required all the same.

Investment in own development.

At the beginning of the new millennium, the management of Otto Hofstetter AG resolved to find or develop a coating which featured all the positive characteristics of the TiN coating, but which was also much more service friendly. An agreement was reached with one of the world's leading coating enterprises to develop a new process within a year. This target was reached in 2003. Otto Hofstetter AG was able to present its own newly developed coating under the name of OH-Xtreme® and use it in its moulds.

High versatility.

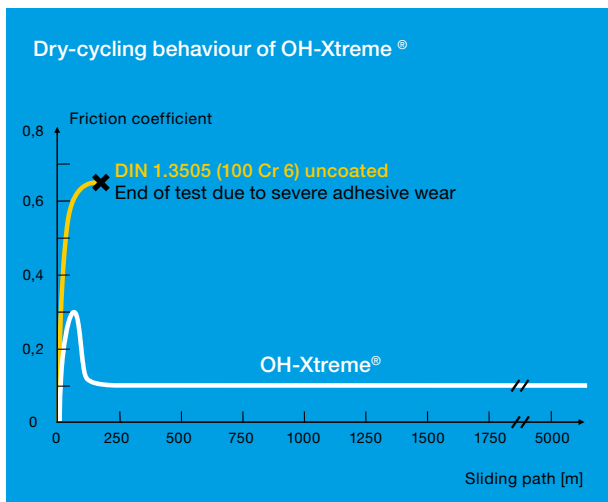
It was the coating of sliding surfaces with OH-Xtreme® which now enabled the production of high-grade plastic articles, for example preforms and other foodstuffs packaging, without resorting to lubricants. OH-Xtreme® combines outstanding sliding properties with high hardness. The special coating protects against seizing and fretting corrosion, and allows the dry running of moving parts such as ejector pins, slides and cores. OH-Xtreme® is an amorphous metal-carbon layer which can be identified by its dark colour.

Efficient and functional protection.

Thanks to their multi-lamellar structure, phases rich in tungsten carbide and rich in carbon alternate within a few layers of atoms. This results in a very low dry-friction value of 0.1 to 0.2, which has a very favourable influence on friction behaviour. Wear and fretting corrosion are almost completely eliminated. Ideally suited to injection moulding processes, the layer prevents the seizing of highly strained components and guarantees impeccable performance even in dry operation.

High output confirmed.

With the first 72-cavity mould delivered featuring the OH-Xtreme® coating, more than 15 million cycles were run straight off without time-consuming maintenance work and without evidence of wear in the coated mould parts. Several hundred moulds have been delivered to customers all over the world since Otto Hofstetter AG introduced this technology. A great deal of positive feedback confirms the generally pleasing results established with the first mould.



After a short increase – the running-in – the dry-friction value is reduced to a minimum thanks to the coating of the parts with OH-Xtreme®.